



Grid-Scale BESS: Technology, Markets, and Project Finance



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This two-day course provides a comprehensive and practice-oriented introduction to grid-scale Battery Energy Storage Systems (BESS) with a clear focus on front-of-the-meter applications. It begins by situating BESS within the wider electricity value chain, unpacking how grid structures, wholesale markets, and EV-driven demand growth are reshaping the role of storage in modern power systems. Participants are guided through the global market landscape for stationary storage, including the main regional trends, deployment drivers, and regulatory signals that underpin project pipelines. Building on this context, the course explores energy storage technologies with an emphasis on lithium-ion, and in particular LFP chemistries, highlighting performance characteristics, safety considerations, commercial maturity, and future technology trajectories. The first day also addresses cost-curve dynamics, long-term CAPEX/OPEX structures, and the realities of global supply chains, including procurement strategies, vendor qualification, and exposure to raw material price volatility and geopolitical risk. The second day then transitions to the commercial and financial dimensions, covering grid storage monetisation (energy arbitrage, ancillary services, capacity markets), value stacking strategies, emerging models such as Battery-as-a-Service principles, and the fundamentals of project evaluation, asset performance, PPAs, and circular economy pathways for second-life and recycling. By linking market context, technology, and finance, the course offers an integrated view of how BESS projects are conceived, financed, and operated in practice.



LOCATION

Rio de Janeiro

London

Singapore

LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future



COURSE OBJECTIVES

This course is tailored for a mixed audience of technical and financial professionals who are either entering the BESS space or seeking to deepen their understanding of grid-scale storage projects. It is highly relevant for power system and electrical engineers, project engineers, and technical advisors involved in grid connection studies, system design, and performance evaluation of battery assets. It also targets project developers, asset managers, and commercial managers responsible for shaping storage business cases, crafting offtake structures, and overseeing operational performance. On the finance side, the course serves investment analysts, project finance practitioners, corporate finance teams, and infrastructure investors who need to interpret technical assumptions, stress-test project models, and understand how operating profiles, degradation, warranties, and regulatory frameworks influence risk and returns. In addition, policymakers, regulators, and utility planners will benefit from a structured view of how BESS assets interact with markets, policy incentives, and ESG expectations. While the course assumes a basic familiarity with power or infrastructure projects, it does not require deep prior expertise in batteries, making it suitable for cross-functional teams who want to build a shared language around BESS.



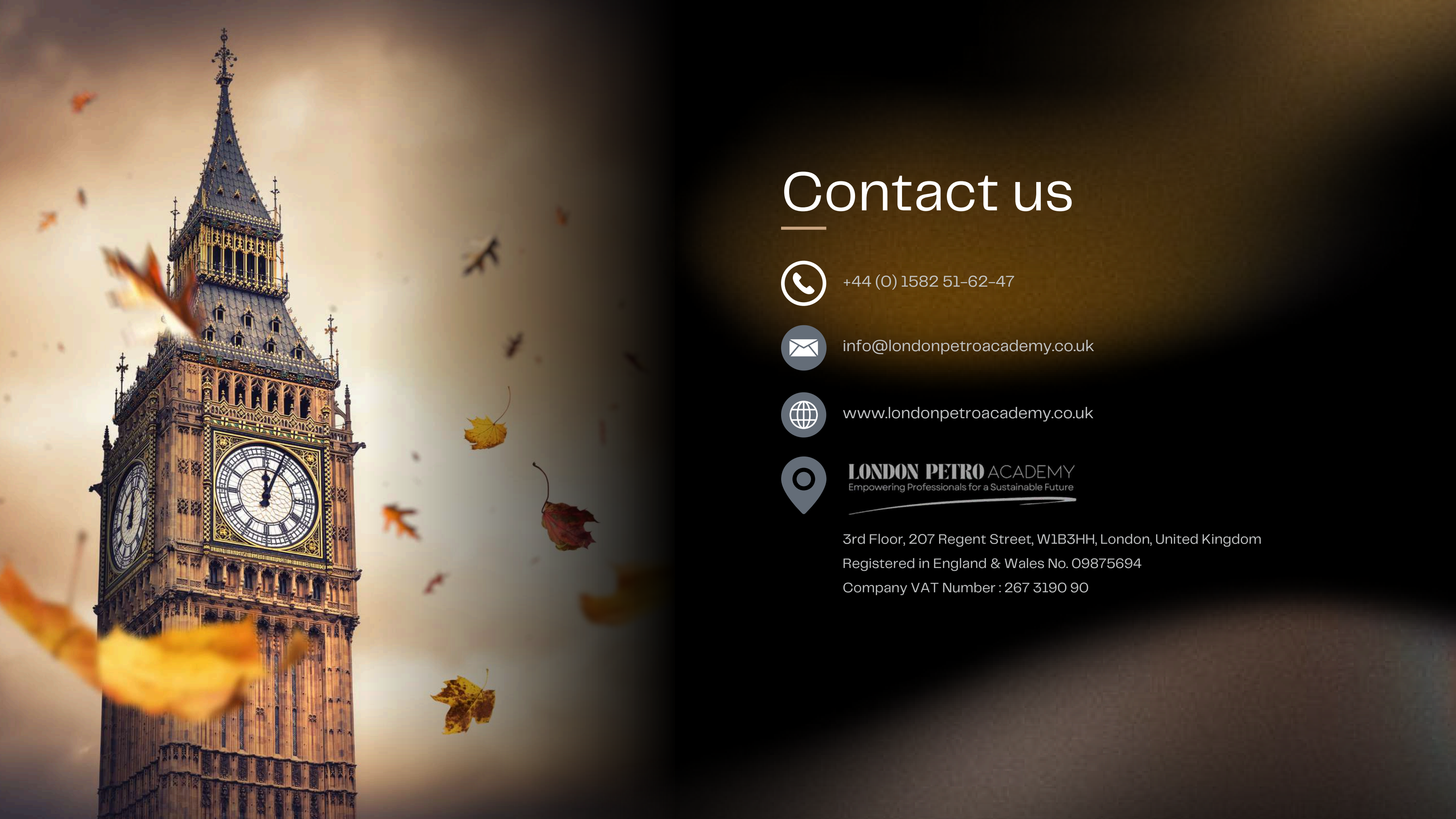
The objectives of the course are to equip participants with both the technical literacy and financial toolkit needed to critically evaluate grid-scale BESS opportunities and risks. By the end of the programme, participants will be able to interpret the global BESS market landscape and identify the main front-of-the-meter use cases, revenue mechanisms, and regulatory enablers that drive project viability. They will be able to compare key storage technologies and lithium-ion chemistries, explain how parameters such as round-trip efficiency, depth of discharge, cycling profiles, and degradation trajectories feed into project sizing, warranty structures, and lifetime performance expectations. On the financial side, participants will gain confidence in reading and challenging project evaluations built around CAPEX, OPEX, NPV, and IRR, understanding how different operating strategies, contract structures, and financing options influence cash flows and bankability. They will also learn to recognise the importance of supply chain robustness, raw material exposure, and vendor selection in risk assessments, and to position projects within ESG-aligned policy and reporting frameworks. Finally, participants will come away with a clear conceptual understanding of second-life applications, recycling technologies, and circular economy models for batteries, enabling more informed decisions across the full life cycle of BESS assets.

Day 1

- Understanding The Grid & Global Market Landscape
- Stationary Front-of-the-Meter (FTM) battery market landscape
- High-level overview of EV sector dynamics and market growth drivers
- Electricity Market Structures Overview
- Energy Storage & Battery Technology Trajectories
- Overview and comparison of electrochemical and alternative storage technologies
- Commercial maturity and technology roadmaps
- Deep dive into Lithium-Ion battery variants with an emphasis on LFP chemistry
- Cost-curve forecasting: Long-term CAPEX/OPEX structures
- Supply Chain Structures & Procurement Risk
- Global sourcing and logistics strategy
- High-level summary of raw material economics
- Strategic risk assessment: Geopolitical disruptions and supply constraints
- Vendor qualification and procurement planning

Day 2

- Value & Storage Monetization Business Models
- Grid storage monetization: Energy arbitrage, frequency regulation, and capacity markets
- Value stacking and revenue streams for front-of-the-meter applications
- Conceptual overview of emerging models (e.g., Battery-as-a-Service / BaaS principles) (Note: Discussion focuses industry principles; real corporate business models are excluded)
- Finance, Investment & Project Evaluation
- Operational parameters impacting project finance (DoD, efficiency, lifetime)
- Project investment evaluation frameworks: CAPEX, OPEX, NPV, and IRR
- Standard corporate and project finance structures for battery deployment
- Performance, Asset Operations & Circular Economy
- Performance guarantees, warranties, and degradation limits
- Overview of Power Purchase Agreement (PPA) structures
- Core principles of asset management: Fundamentals of cost-efficiency and performance metrics (Note: Concrete asset management strategies are excluded)
- Circular economy: Second-life battery applications and commercial viability
- Focus on Li-Ion battery recycling technologies and material recovery pathways
- Risk, regulation, and general guidelines for ESG-aligned policy frameworks



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